

Pollinator Module

Retraining a model

Choose which model you want to retrain. Click **(1)** to train YOLO

New training job

Pick a model, choose data, and start.

1. PICK A MODEL TO RETRAIN ⓘ

Full-Image Detection

Retrain independently of the motion-based pipeline.

- 1
- ☒ **YOLO** (Full-Image Detection) **YOLO-v1**
Scans every image directly to find and label insects — no motion required.
mAP50 — 8 samples available (+8 new since active)

Motion-Based Detection

Two models that work in sequence — retrain each separately.

- ☐ **Binary Classifier** (Insect Detection)
Step 1 — checks each moving object and decides: is this an insect or just background noise?
- ☐ **Group Classifier** (Insect Classification)
Step 2 — takes confirmed insects from Step 1 and classifies them as fly, bumblebee, butterfly, or other.

Choose which model to train from (2). Click (3) to browse images/crops. Tick/untick (4) to exclude/include in training. (5) Exclude or include all images. Untick (6) to only use uploaded dataset. (7) Upload images and labels. (8) Choose which classes should be included in training.

2. CHOOSE TRAINING DATA

Base model **YOLO-v1 (active)**

6 ☒ 8 samples in pool
8 from review · 0 from uploaded dataset

3 Browse pool ->

• 10 fly

YOLO dataset (.zip) ⓘ

7 

Drop a YOLO dataset .zip, or click to browse
Requires images/, labels/, and data.yaml. Classes are remapped to this model.

8 Classes to include ⓘ

☒ Bumblebee ☒ Fly ☒ Butterfly ☒ Other

Filter: **All** Fly **4** Include all · Exclude all

Include	Image	Detections	Classes
5 ☒	Ara1_Ara1_p1_101_WSCT_WSCT9771_NvYuyI1.JPG	1	fly
☒	Sia_Sia_p1_103_WSCT_WSCT0115_myMzh8c.JPG	1	fly
☒	Sia_Sia_p1_103_WSCT_WSCT0116_ERP1XZH.JPG	1	fly
☒	Sia_Sia_p1_103_WSCT_WSCT0118_3HV7TCO.JPG	2	fly
☒	Sia_Sia_p1_103_WSCT_WSCT1285_uVcG6M5.JPG	1	fly
☒	Sia_Sia_p1_103_WSCT_WSCT1286_1848cnn.JPG	1	fly
☒	Sia_Sia_p1_103_WSCT_WSCT1293_0B6AmVx.JPG	2	fly
☒	Sia_Sia_p1_103_WSCT_WSCT1301_FiUkctj.JPG	1	fly

Click **(9)** to select/unselect for training. Blue underline means fly and gray means background. Make sure they match what is actually in the crops.

Review training pool Binary Classifier · 107 of 110 included ✕

Filter: **All** Insect Background Include all · Exclude all

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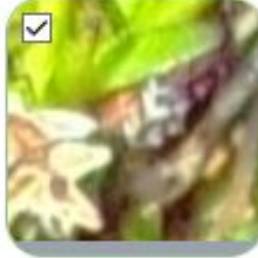
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The model benefits from more data if it is high quality. Review the output thoroughly to ensure the best training data. It is recommended that you don't train with less than 500 images/crops as it can lead to catastrophic degradation of the model. 500 should be view as the minimum amount and further increasing the amount of data will be beneficial.

